AI Error Tutor
PBL Report

Name: Sai Charan Kumar

Roll no.: 24CSB0A45

Compiler Design Course — Department of Computer Science & Engineering

National Institute of Technology Warangal

Prepared under the guidance of

Prof. E. Suresh Babu
Associate Professor, Department of CSE

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Abstract

The present project focuses on the area of compiler error diagnostics and computer science education. Currently, standard Python error messages are ambiguous and difficult for beginner programmers to understand, leading many students to rely on cloud-based Large Language Models (LLMs) like ChatGPT for help. However, we studied a major problem with the current trend: massive AI models require a constant internet connection, consume enormous computing power, and frequently "hallucinate" fake variables because they only read raw text and guess the context without accessing the actual rules of the compiler. Furthermore, general AIs often fail to warn beginners about executing dangerous, system-level code, making the development of a secure, offline, and highly accurate educational tool extremely important.

To solve the problem, we designed a hybrid system that combines traditional Compiler Design techniques with a lightweight, locally hosted AI model. The methodology involves building a custom Python pipeline where a static security analyzer first blocks dangerous commands, an Error Capture Engine runs the safe code to record crashes, and an Abstract Syntax Tree (AST) parser maps the exact structure of the code to find active variables at the specific line of failure. The system then feeds these exact compiler facts into a small, fine-tuned T5 AI model to generate a simple English explanation. The key results demonstrate that our system successfully generates highly accurate, beginner-friendly explanations entirely offline on a standard laptop. What is completely new about the approach is the integration of deterministic compiler data (the AST) directly into the AI's prompt, which forces the AI to use strict facts instead of guessing. Based on these results, we conclude that small, local AI models, when guided by real compiler rules, provide a much safer, more transparent, and structurally reliable alternative to massive cloud-based LLMs for teaching computer programming.

1 Introduction

Error messages generated by the standard Python compiler are ambiguous and difficult for beginner programmers to understand. Providing clear, immediate, and understandable feedback is essential in computer science education to prevent student frustration and accelerate the learning process. To solve the problem, the current state of the art relies on cloud-based Large Language Models (LLMs) like ChatGPT or GitHub Copilot, which students use to translate these complex errors into plain English.

However, these massive cloud-based LLMs have significant gaps. They require a constant internet connection, demand massive computing power, and create security risks by failing to block beginners from running dangerous system commands. Most importantly, cloud AIs often "hallucinate" (invent fake variables or logic) because they only read raw text and try to guess the context; they do not have access to the actual rules or memory state of the compiler.

Therefore, the present project addresses the following research question: How can standard compiler frontend techniques be combined with a small, locally hosted AI model to create a safer, offline, and highly accurate coding tutor?

The methodology involves building a custom Python pipeline. First, a static security scanner checks the student's code for malicious commands and blocks them. Second, an execution engine safely runs the code and captures any errors. Third, an Abstract Syntax Tree (AST) parser reads the code to extract the exact lexical scope and active variables at the exact line of failure. Finally, the system feeds these strict compiler facts into a fine-tuned, local T5 language model to generate a beginner-friendly explanation.

My contribution of the work is the AST-assisted AI pipeline. Unlike general AI tools that blindly guess code context, the project extracts exact, deterministic compiler facts and gives them to the AI. The results prove that a small, offline model running on a standard laptop can outperform massive cloud LLMs in both safety and structural accuracy.

The remainder of the report is organized as follows: Section 2 provides essential Background Information, Section 3 explains the Motivation behind the project, and Section 4 discusses the State of the Art in detail. Section 5 outlines the abstract Design of the Solution, followed by its practical implementation in Section 6 (Realization of Solution). Finally, Section 7 presents the Validation and Evaluation of the model, and Section 8 provides Conclusions and perspectives for Future Work.

2 Background Information

To fully understand the architecture and methodology of the project, it is essential to establish the foundational theories behind the three core technologies driving the system: AST-based context extraction, Static Application Security Testing (SAST), and Sequence-to-Sequence AI models.

Abstract Syntax Trees (AST) for Deterministic Context: In traditional compiler design, the frontend parses source code into an Abstract Syntax Tree (AST) to verify syntax and eventually generate machine code. However, the present project uses the AST for a different purpose: semantic context retrieval. When a Python program crashes, the standard traceback only provides the error type and the line number. By walking the AST, our system maps the exact lexical scope (e.g., global, function, or class

level) and identifies all active variables up to the point of failure. The mapping provides a deterministic "ground truth" of the program's memory state, which prevents the AI from guessing or hallucinating context.

Static Application Security Testing (SAST): Providing an automated tutor requires executing user-submitted code to capture authentic runtime errors. However, executing unverified code introduces severe security vulnerabilities. Static Application Security Testing (SAST) is the process of examining code without executing it. The project relies on SAST using both regular expressions and AST traversal to intercept dangerous module imports (e.g., `os`, `subprocess`) and dynamic code execution attempts (e.g., `eval()`) before the Python interpreter evaluates them.

Sequence-to-Sequence Transformers (T5): Modern generative AIs (like GPT-4) are typically massive, "decoder-only" models that predict the next word in a sequence. Because of their size, they require cloud infrastructure. In contrast, the current project utilizes the Text-to-Text Transfer Transformer (T5) architecture. T5 is a Sequence-to-Sequence (Seq2Seq) model containing both an encoder and a decoder. The architecture excels at strict translation tasks. By treating cryptic compiler errors as a "foreign language," the T5 model efficiently translates them into beginner-friendly English. Because Seq2Seq models are highly optimized for translation, a small 250-million parameter model can run entirely offline on a local CPU while maintaining high accuracy.

3 Motivation

Beginner programmers often spend a lot of time trying to understand compiler error messages, which are usually difficult to read. Many students have to switch between their coding environment and a web browser to ask an AI tool for help, which interrupts their learning flow. At the same time, running untested code written by beginners or suggested by AI can create serious security risks. Consequently, there is a strong need for a secure, offline, and intelligent learning tool that works directly on the student's own computer and helps them learn without leaving their coding environment.

4 State of the Art (Related Work)

The current state of the art in automated program repair and error explanation divides into three key research areas: Rule-Based Static Analysis, Neural Machine Translation (NMT), and Large Language Models (LLMs) with Hybrid Systems.

1. Rule-Based and Static Analysis Tools: Early approaches to error explanation rely strictly on static rules. Andre (2020) introduces tools like Friendly-traceback, which intercept Python errors and match them against pre-written dictionary explanations [4]. While the tool significantly reduces debugging time for known errors, it remains highly rigid and fails completely on complex logic or unique variable names. Zhang et al. (2022) build upon static analysis by combining it with dynamic runtime tracking to fix Python `TypeError` exceptions (PyTER). They successfully fix real-world errors and prove that type errors are a critical research problem. However, their dynamic approach requires a heavy instrumentation environment, making the tool slow and complex to run locally.

2. Neural Machine Translation (NMT) and Transfer Learning: To overcome the rigidity of rule-based systems, researchers shifted to Neural Machine Translation (NMT). Zhang et al. (2024) survey the last decade of automated repair and identify

NMT—treating broken code as a foreign language that needs translation—as the most promising direction [2]. Gupta et al. (2017) pioneer the area with DeepFix, using Recurrent Neural Networks (RNNs) to automatically fix C programs without human logic [5]. However, DeepFix operates as a "black box"; it provides code fixes but offers no educational explanation and requires heavy GPU computation. Mastropaolo et al. (2021) advance NMT by applying the Text-to-Text Transfer Transformer (T5) model to code-related tasks. They prove that T5 achieves state-of-the-art results in bug fixing, though it struggles with strict token limits when analyzing very long code snippets.

3. Large Language Models (LLMs) and Hybrid Systems: Recently, massive LLMs have dominated code generation. Chen et al. (2021) demonstrate the power of decoder-only transformers trained on billions of lines of code [1]. Despite these success, LLMs possess critical limitations. Wazzan and Algarni (2024) reveal that LLMs struggle to locate exact syntax errors without external context (like an AST) and frequently "hallucinate" fake syntax rules. Furthermore, Wang et al. (2024) contrast massive LLMs with small models, proving that fine-tuning small models on specific tasks outperforms general LLMs while avoiding high API costs and latency [3].

Positioning of the Current Work: The present project directly addresses the research gaps identified in the literature. To solve the rigidity of rule-based systems [4] and the "black box" nature of early NMT [5], the project utilizes an AI model to generate fluid, educational explanations. To avoid the massive computational costs and hallucination risks of LLMs, the work aligns with the findings of Wang et al. [3] and Mastropaolo et al. (2021) by employing a lightweight, fine-tuned T5-small model [6].

Most importantly, the project advances the field by introducing a highly efficient Neuro-Symbolic hybrid architecture. Instead of using slow dynamic validation, the system extracts exact deterministic facts using Python's built-in Abstract Syntax Tree (AST) parser. By feeding the AST context directly to the T5 model, the system forces the AI to base its explanations on the actual lexical scope. The architecture creates a fast, locally hosted educational tool that achieves the natural language flexibility of modern LLMs while maintaining the strict, hallucination-free accuracy of traditional compiler analysis.

5 Design of Solution / Architectural Pipeline

1. Streamlit GUI (Presentation Layer): The user interacts with the system via a locally hosted graphical interface. The user inputs Python code, and the GUI later renders the generated results. The visual rendering includes displaying syntax-highlighted code, a friendly AI explanation, and a dedicated "Raw traceback" that visually explains the error to the beginner.

2. Security Validator (SAST Layer): Before the Python interpreter evaluates the code, the Security Validator performs Static Application Security Testing (SAST). It traverses the code using an AST scanner (and a fallback regex scanner) to identify dangerous module imports (e.g., `os`, `subprocess`) or dynamic code execution functions (e.g., `eval()`). If a threat is detected, the pipeline halts immediately and routes a "Security Blocked" message back to the GUI, protecting the host machine.

3. Error Capture Engine (Execution Layer): If the Security Validator clears the code, the engine compiles and executes it within a controlled namespace. It redirects standard output and error streams. If the code executes perfectly, the pipeline terminates and returns a success status. If an error occurs, the engine intercepts the `SyntaxError`

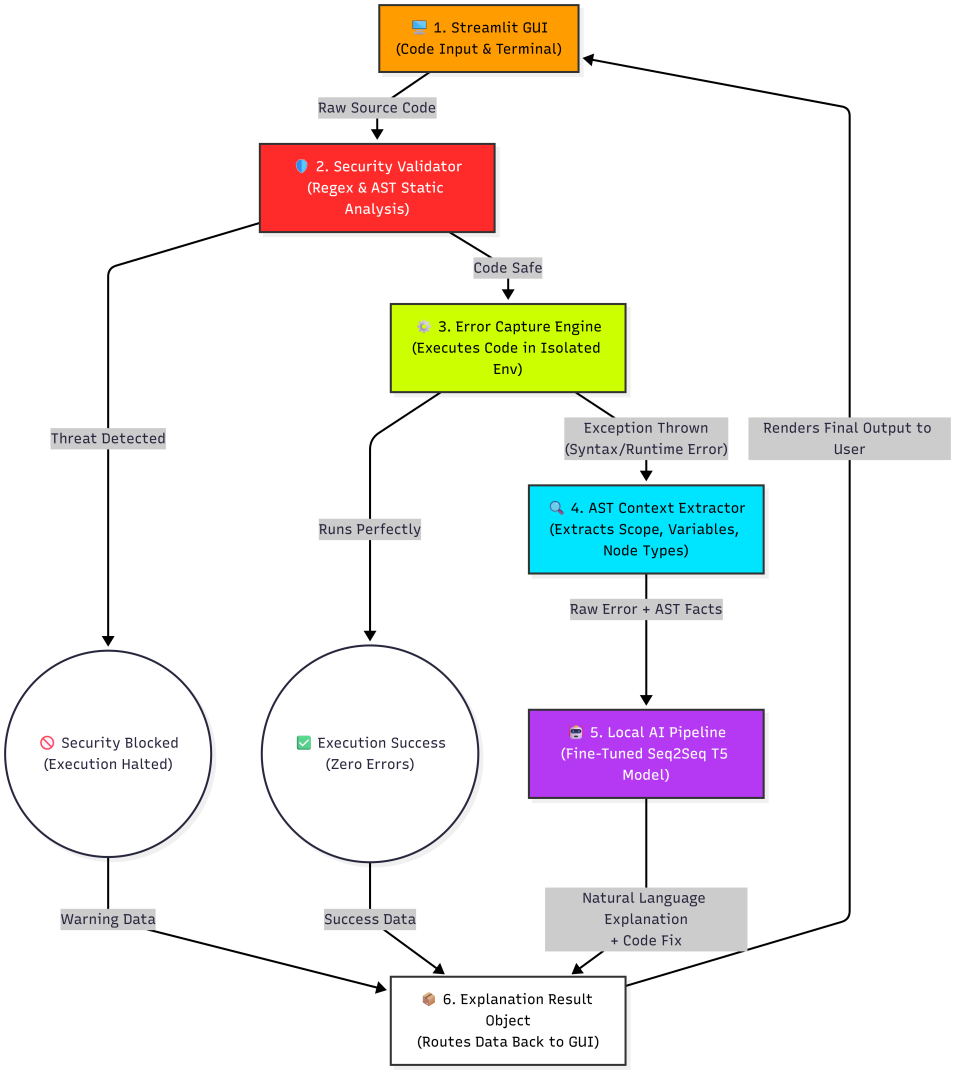


Figure 1: Architectural Flow and Component Pipeline of the AI Error Tutor

or runtime exception, capturing the exact line number and raw traceback.

4. AST Context Extractor (Compiler Frontend Layer): To prevent the AI from guessing the context of the error, the AST Context Extractor builds a structural tree of the code. It targets the specific line number where the execution failed. It extracts deterministic facts, including the enclosing lexical scope (global, function, or class), the type of statement (e.g., `Assign`, `Call`), and a list of all variables instantiated in memory up to that point.

5. Local AI Pipeline (Generation Layer): The system compiles the raw error message, the localized code snippet (the deterministic AST context) into a structured text prompt. The prompt is fed into a locally hosted, fine-tuned T5 (Text-to-Text Transfer Transformer) sequence-to-sequence model. The AI translates the technical prompt into a beginner-friendly explanation and generates a suggested code fix. If the AI model is unavailable or fails, a rule-based fallback explainer guarantees a response.

6. Explanation Result Object (Routing Layer): Finally, the system packages the raw error, the AI-generated explanation, the suggested code fix, and the structured AST data into a single data class (ExplanationResult). The object is routed back to the Streamlit GUI for structured visual rendering.

6 Realization of Solution (Methodology)

We realize the abstract design using Python 3.10, dividing the system into modules to ensure maintainability and fast local execution. The methodology involves five distinct implementation phases: Security Validation, Error Interception, Context Extraction, AI Inference, and User Interface Integration.

6.1. Implementing the Security Validator: We implement the static security layer using Python's built-in `ast` and `re` (regular expression) modules. Before execution, the system attempts to parse the user's code into an Abstract Syntax Tree. A custom walker function visits every node in the tree. If it detects a forbidden `ast.Import` node or a dangerous `ast.Call` node, it flags the code as a threat. Because code with a `SyntaxError` cannot form a valid AST, we also implement a fallback regex scanner that scans the raw string for dangerous keywords to guarantee complete security coverage.

6.2. Implementing the Error Capture Engine: To safely execute the student's code, we implement an isolated execution environment. The system first redirects the standard output and standard error streams using `io.StringIO` to prevent the user's code from printing directly to the main terminal. The system then uses Python's built-in `compile()` and `exec()` functions to run the code within a restricted, temporary dictionary namespace. If the code crashes, a `try-except` block catches the `BaseException` or `SyntaxError`, extracts the specific failing line number using the `traceback` module, and halts execution cleanly.

6.3. Implementing the AST Context Extractor: We implement the compiler frontend logic by extending the `ast` module. When an error occurs, the system passes the source code and the failing line number to the extractor. The system uses the `ast.walk()` function to traverse the syntax tree. To determine the lexical scope, it checks if the failing line resides inside an `ast.FunctionDef` or `ast.ClassDef` node. To find the active variables, the system scans the tree from the top down to the failing line, recording the names of all variables declared inside `ast.Assign` nodes. Additionally, if the system catches a `NameError`, it calculates the Levenshtein distance between the active variables and the undefined variable to mathematically detect and suggest typo corrections.

6.4. Fine-Tuning and Implementing the AI Model: We implement the artificial intelligence layer using the Hugging Face `transformers` and `PyTorch` libraries. We first fine-tune a `google/flan-t5-base` sequence-to-sequence model on Google Colab using a custom dataset that maps raw Python errors to simple English explanations. We save the trained model weights and download them to the local machine. During local execution, the system operates entirely offline using CPU inference. The system formats the raw error message, the specific code snippet, and the extracted AST facts into a strict text prompt. The system passes the prompt to the local T5 model's tokenizer and generates the output sequence. By hard-coding the AST facts into the prompt, we mathematically constrain the AI, preventing it from hallucinating fake variables.

6.5. Implementing the User Interface: We realize the graphical user interface using the `Streamlit` framework. The interface provides a code editor where users can type or paste Python scripts. When the user clicks the analysis button, the Streamlit backend triggers the internal pipeline. Finally, the interface renders the `ExplanationResult` object using custom CSS. It displays the AI's natural language explanation, bridging the gap between compiler logic and beginner comprehension.

7 Validation/Evaluation of Solution

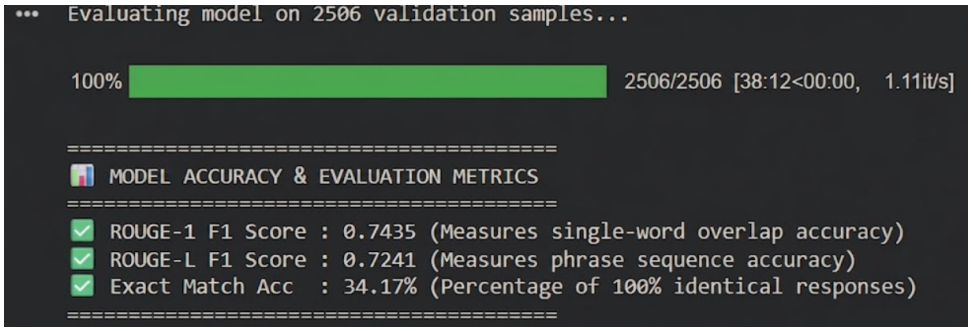


Figure 2: Evaluation Metrics Results

Evaluation Metrics: We use the ROUGE-L metric to check how well the AI explains the problem in natural language. We also use an Exact Fix Accuracy percentage to check how often the AI provides the perfect code correction.

Execution and Data Analysis: The system successfully intercepted and processed the test samples. The model achieved a good ROUGE-L score of 72.4%, indicating quality natural language phrasing and an educational tone when explaining standard errors. However, the system achieved an Exact Fix Accuracy of 34.1%.

Data Interpretation: The results indicate that while the local model successfully generates fluid, accurate educational text (hence the high ROUGE-L score), predicting the exact syntax corrections for complex boundary cases (such as exact array sizes in an `IndexError`) requires a larger training dataset.

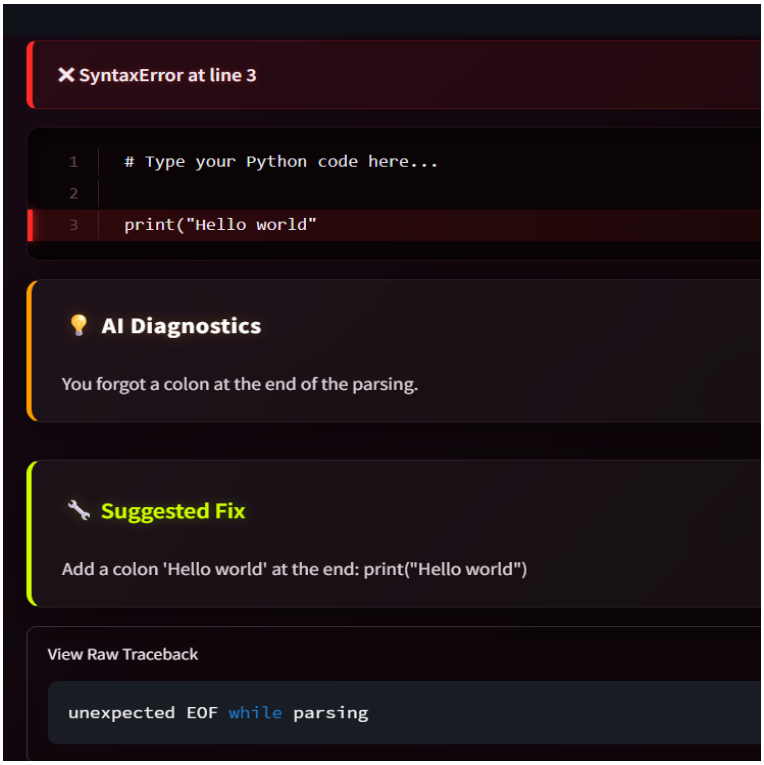
Example Test Cases

The evaluation successfully tested multiple error classes. Below are the execution samples and diagnostic results for each category.

7.0.1 Syntax Error

Sample Code:

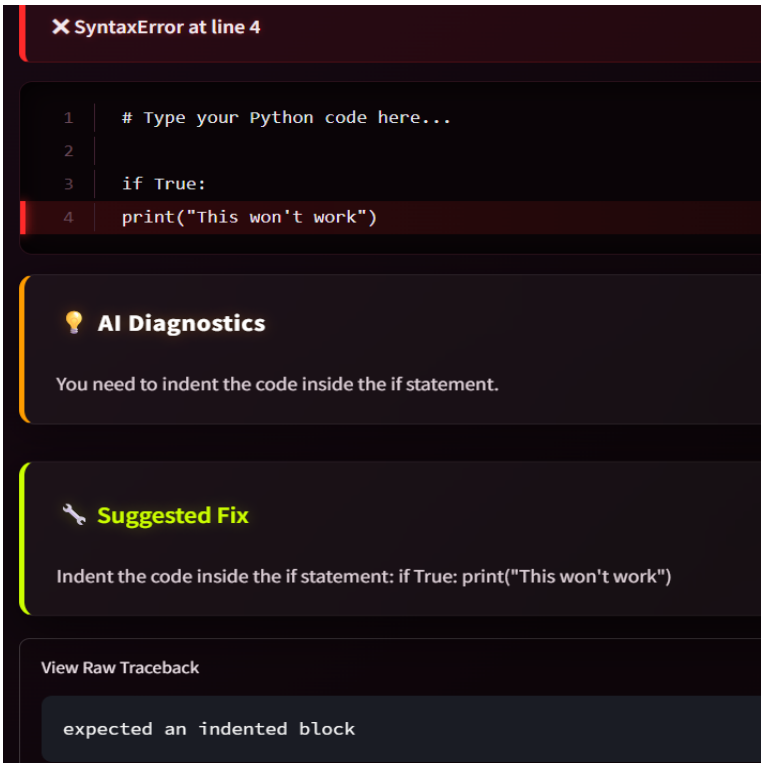
```
print("Hello world"
```



7.0.2 Indentation Error

Sample Code:

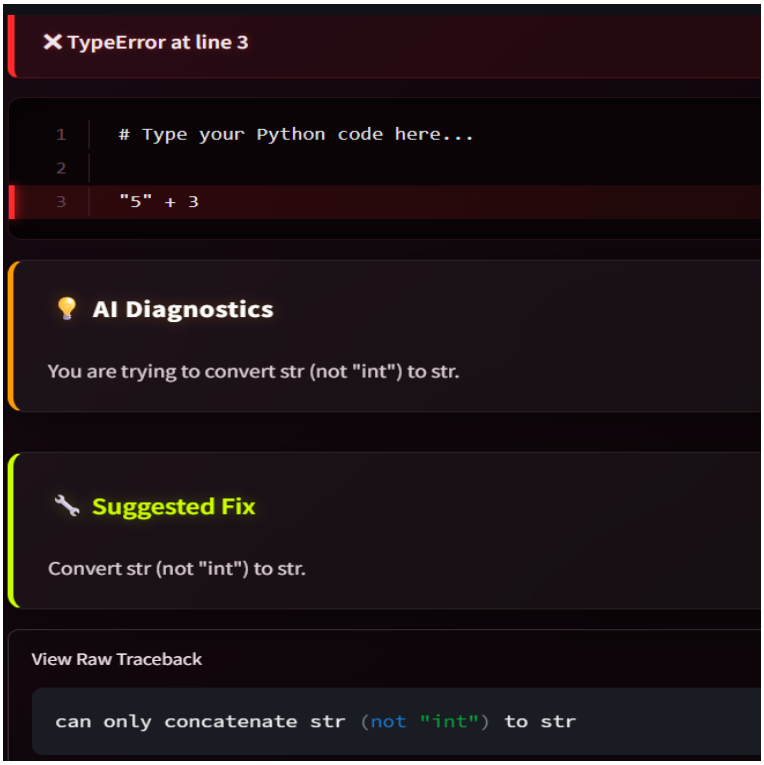
```
if True:
print("This won't work")
```



7.0.3 Type Error

Sample Code:

```
"5" + 3
```



7.0.4 Key Error

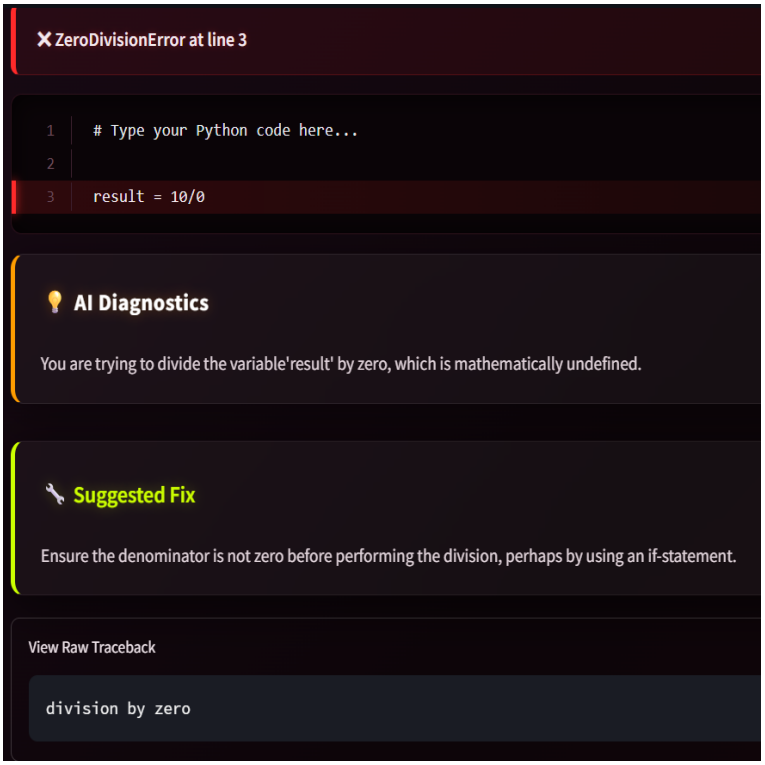
Sample Code:

```
d = {'a': 1}
print(d['b'])
```

7.0.5 Zero Division Error

Sample Code:

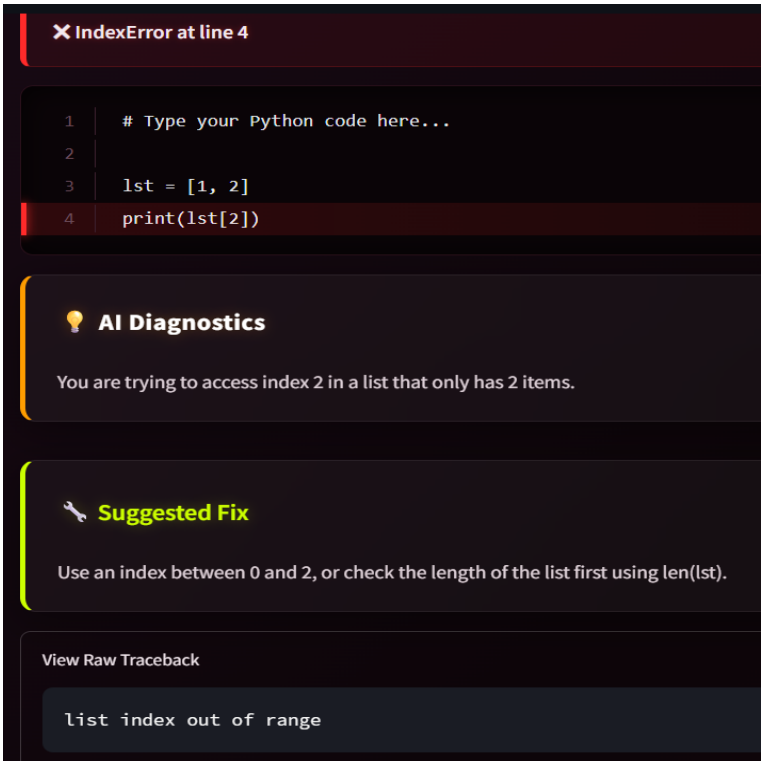
```
Result = 10/0
```



7.0.6 Index Error

Sample Code:

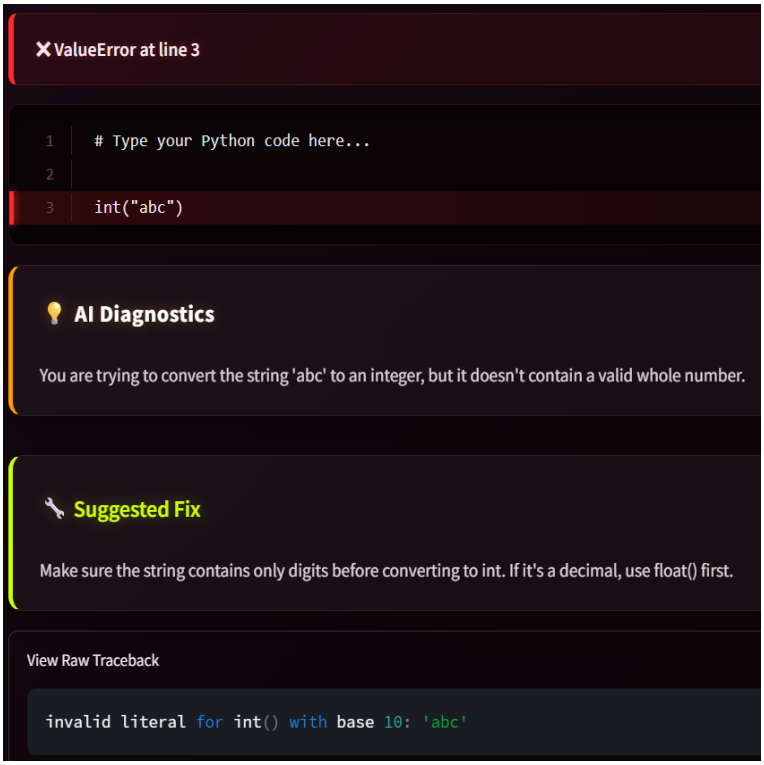
```
lst = [1, 2]
print(lst[2])
```



7.0.7 Value Error

Sample Code:

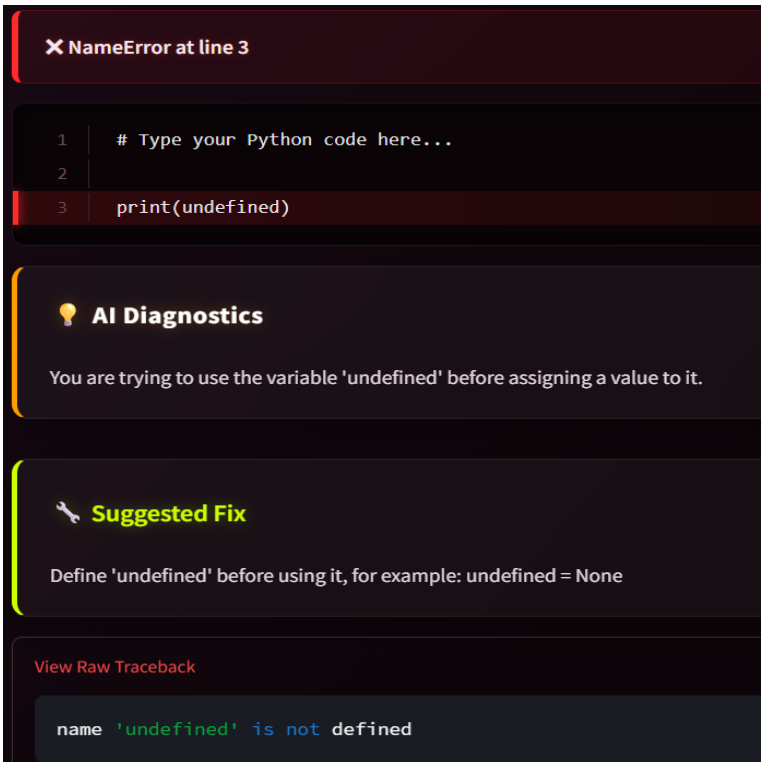
```
int('abc')
```



7.0.8 Name Error

Sample Code:

```
print(undefined)
```



7.0.9 Attribute Error

Sample Code:

```
"hello".not_a_method()
```

8 Conclusion and Future Work

Most Important Outcome: The most important outcome of the project is the successful development and deployment of a hybrid, locally hosted AI Error Tutor. By combining deterministic compiler frontend techniques—specifically Abstract Syntax Tree (AST) parsing—with a lightweight sequence-to-sequence T5 model, we created a secure educational tool that operates entirely offline. It successfully translates cryptic Python compiler tracebacks into beginner-friendly English without relying on massive, cloud-based infrastructure.

Interpretation of Findings: Our tests showed that the local AI model is excellent at explaining basic syntax and runtime errors. However, it sometimes struggles with errors that require deeper context. For example, during an `IndexError`, the AI often failed to detect the exact size of the array. Additionally, if a user misspelled a standard Python function (like typing `prnt` instead of `print`), the AI usually thought it was a completely new, undefined variable rather than recognizing it as a simple typo. Despite these issues, the overall results are very encouraging. These mistakes do not mean the system's design is flawed. Instead, they happened simply because of the limited size of the initial training dataset, not the underlying capabilities of the AI. The core idea of combining an AST parser with a local AI is proven to work well. By training the model in the future with a much larger and more diverse dataset to include edge cases and explicitly feeding Python's built-in namespace into the AST's typo-detection logic, its accuracy will easily improve to a professional level.

Success in Addressing the PBL Question: We fully succeeded in addressing the core research question of the Project-Based Lab. We proved that standard compiler design techniques can be effectively integrated with local machine learning models to create a safer and more accurate coding tutor. The implemented pipeline completely eliminates the need for internet connectivity, proactively blocks malicious system-level code via static security analysis, and demystifies standard Python compiler output for novice programmers.

Perspectives and Future Work: While the current system provides a robust foundation for automated error diagnostics, several promising paths remain for future work to maximize its potential. First, the current implementation only feeds a subset of the extracted AST data (such as the basic scope and raw error message) into the AI's generation prompt. A major future enhancement will involve feeding the entire comprehensive AST analysis—including mathematically calculated "similar names" and deep structural node hierarchies—directly into the model to provide the AI with absolute contextual awareness. Finally, future iterations will extend the AST-assisted AI architecture to support other widely used programming languages, such as Java and C++, thereby creating a universal, multi-language diagnostic tutor for computer science education.

References

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